

DATA ANALYSIS · E-COMMERCE

Did Free Shipping Increase Profit?
Shipping Cost Analysis — January to September 2024

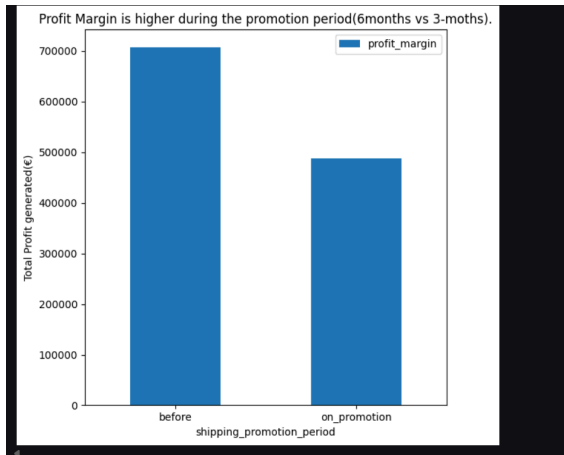
Period: Jan 2024 – Sep 2024
Baseline: Jan – Jun (6 months)
Promotion: Jul – Sep (3 months)

KEY METRICS

Net Promotion Profit	Shipping Absorbed	Revenue (3 mo vs 6 mo)
€422,760	€36,610	78.2%

KEY FINDINGS

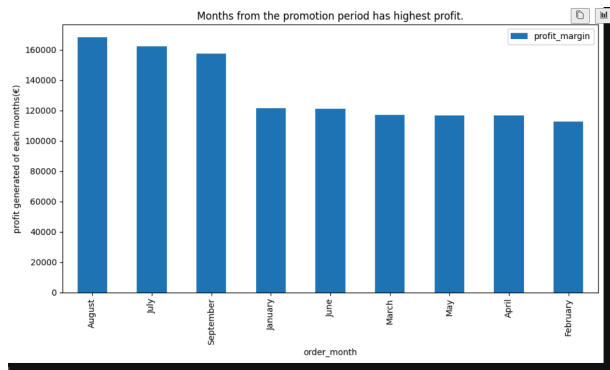
- Revenue surged. The 3-month promotion period captured 69% of the full 6-month baseline or revenue— roughly double the monthly rate.



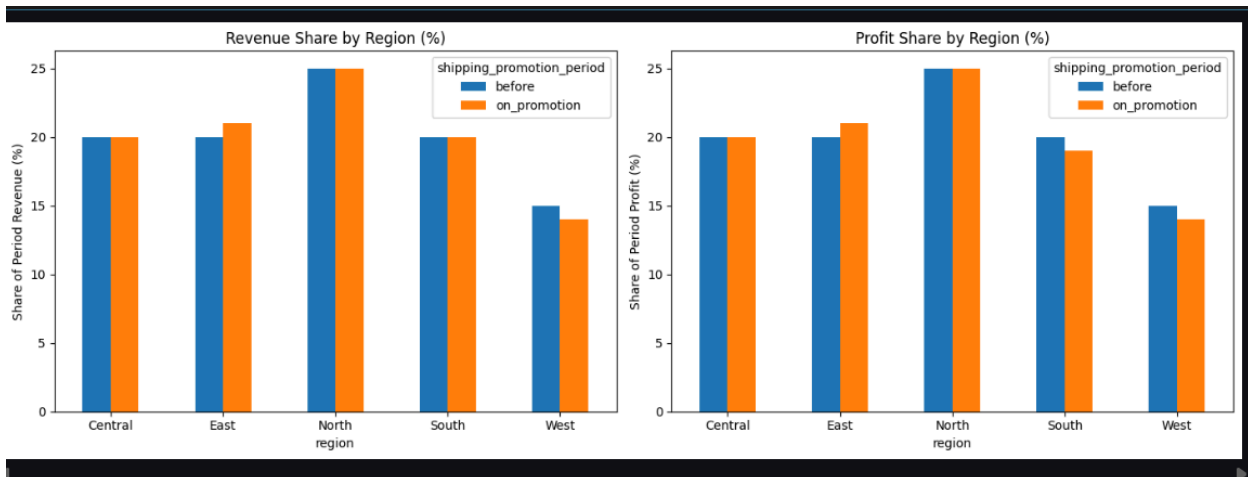
- Customers spent more. Spending tier analysis shows a measurable shift toward the €50+ bracket, suggesting customers actively topped up carts to qualify for free shipping.



- Profit margin grew per month. Months of the promotion period have the highest profit margin.



- All regions improved. Revenue and profit grew across all 5 regions (North, South, East, West, Central) during the promotion period.
 - All regions have generated almost similar profit and revenue. So the company needs to equally address customers from all regions.



- Shipping subsidy well-covered. The €36,610 in absorbed shipping costs represent only 8.7% of the net promotion profit

RECOMMENDATION

VERDICT: Continue free shipping above €50.

- The promotion generated strong growth in order volume, average order value, and profit margin. The cost of absorbing shipping fees is comfortably offset by incremental revenue.
- There is no evidence the company simply traded shipping income for equivalent product sales. The net result is a positive gain.

Caveat.

- Taking the promotion is applied during the **summer months(busy market in Europe)**, maybe the raise in profitability is raised from the period.

ASSUMPTIONS & LIMITATIONS

- Analysis covers Jan–Sep 2024 only – no full annual baseline available for seasonal control. Comparisons between the 6-month baseline and 3-month promotion periods are normalised per-month where relevant.

- 452 malformed order line records and 220 duplicate line items were excluded; assumed to be random noise.
 - Unit cost values stored with European decimal notation were corrected prior to margin calculations.
-